PRAVEEN KUMAR OJHA

Process & Data Analyst

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MOTIVATION

Data Analyst with 4+ years of experience in Process Mining and Data Engineering at Bosch. Skilled in Celonis optimization, LLM-based conformance checking, and hands-on machine learning projects. Currently blending ML and analytics with process intelligence as part of my master thesis. Seeking roles where I can apply **data-driven thinking** to **optimize** and **automate** business processes.

🤝 *Let's connect if you're solving **complex process challenges** or need sharp, **scalable data analysis expertise**!*

TECHNICAL SKILLS

CORE EXPERTISE

SQL	4/5
Machine Learning	4/5
Streamlit	4/5
Python	3/5
Git	4/5
PyCelonis	4/5

TECHNOLOGY LANDSCAPE

Process Mining: Celonis, pm4py, OCPM, ETL

Optimization: Heuristic Opt., gurobipy, cplex

ML & Viz: scikit-learn, tensorFlow, seaborn

Infra & Web: Flask, Docker, Grafana, Shell, pyodbc

PROCESS KNOWLEDGE

- O2C (Order-to-Cash)
- B2C (Bill-to-Cash)
- P2P (Procure-to-Pay)
- A2A (Affiliate-to-Buisness)

WORK EXPERIENCE

Master Thesis
Bosch
03/2025 - Ongoing — Stuttgart, Germany

- Building an LLM benchmarking pipeline by integrating Signavio and Celonis, automating activity mapping, and visualizing results in Streamlit—aiming to boost evaluation speed by 40% and enabled token-accuracy analysis across use cases.

Data Engineering in Process Mining Intern
Bosch
09/2024 – 02/2025 — Stuttgart, Germany

- Streamlined Celonis platform tasks by scripting Celonis platform tasks (Data Pool backups, job health checks, and adoption metrics), improving task execution time by 60%.
- Validated access rights across PyCelonis and UI, ensuring full alignment with role-based standards and reducing permission discrepancies
- Deployed regex-driven outlier detection for access anomalies with 95% accuracy; automated monthly audit reports via Action Flows, cutting manual effort by 80%.
- Transformed parquet-based P2P datasets into BI-ready activity and KPI views in Microsoft Fabric, accelerating Microsoft Process Mining deployment cycles.
- Designed Streamlit application which automated process conformance checks across Signavio and Celonis; enabled structured activity mapping workflows used in 3+ pilot use cases during the PoC phase.
- Launched a Gamified leader board using LLM-scored commits to boost versioning, increasing developer engagement by 45% and automated weekly adoption tracking via Celonis ML Workbench and Action Flow.
- Documented and standardized test management strategies in Azure DevOps, conducted workshops for 10+ developers, and helped align UAT execution with the TEST MANAGEMENT 4.0 framework.

Working Student
Bosch
11/2023 – 08/2024 — Stuttgart, Germany

- Optimized Celonis transformation queries and data model loading, cutting use case runtime and memory usage by 40%.
- Built automated permission reporting Views from the Knowledge Model, enabling quick detection of false permissions and improving reporting accuracy by 75%.
- Led a 6-day Celonis platform training for cross-functional participants (3 IT, 2 management), ensuring hands-on adoption of process mining best practices.

Senior Associate Consultant
Bosch Global Software Solutions
07/2022 – 08/2023 — Bangalore, India

- Led B2C process performance tuning, reducing runtime from 24 to 10 hours for 32M data through strategic query and memory optimization.
- Won 1st place in internal hackathon, designing innovative P2P views using the Knowledge Model, improving process transparency.
- Conducted a 1-day workshop for India-Germany teams on performance optimization and script versioning best practices, enhancing code quality awareness.

Software Engineer
Bosch Global Software Solutions
07/2020 – 06/2022 — Bangalore, India

I2P (Invoice-to-Pay)

SOFT SKILLS

Agile Methodology

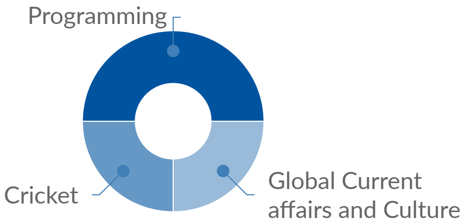
Process Consultation

Problem-Solving

LANGUAGES

Hindi Native
English C1 (Professional)
German A1 (Basic)

INTEREST AREAS



- Orchestrated process flow transition from Sales Order to Purchase Order, covering 3.5M records across 7 systems from a 13-month dataset.
- Developed reusable O2C task templates across 8 regions, reducing manual scripting effort by 30%.
- Migrated 3 on-premise use cases to IBC, coordinating with Celonis Support to ensure seamless transition and functionality.

Associate Software Engineer
Bosch Global Software Solutions
09/2019 – 06/2020 | Bangalore, India

- Supported Celonis AR use case setup, uploading a 30M-record data model from 7 systems via Oracle SQL to SAP HANA, enabling large-scale process analysis.

Data Analyst Intern
Tata Steel
2018 | Jamshedpur, India

- **Developing Employee Turnover Prediction Model -** Developed a predictive model for employee turnover, identifying key attrition factors and helping HR design targeted retention strategies.

EDUCATION

M.Sc. in Data Analytics and Decision Science
RWTH Aachen University | GPA - 1,5
10/2023 – 09/2025

CO_2 EMISSIONS FORECASTING FOR EU COUNTRIES

ML ensemble prediction of EU carbon emissions | Repository

- **Technologies:** seaborn, scikit-learn, xgboost, tensorflow, docker, grafana, postgresql
- Developed a machine learning pipeline to **predict national CO_2 emissions** across EU countries by extending the methodology from a 2024 Scientific Reports paper.
- Benchmarked ML models (GBM, SVR, LSTM, Ensemble) across 10-year horizons in a Dockerized pipeline with Grafana dashboard and PostgreSQL-backed experimentation.

OPTIMIZATION OF TRANSPORT PLANNING AND INBOUND PROCESSING IN PACKAGE CENTER

Routing and scheduling for overnight delivery | Repository

- **Technologies:** gurobipy, pandas, cplex
- This project focuses on optimizing the parcel transportation network of DHL Deutsche Post, which involves overnight delivery across 37 centers.
- The optimization problem involves determining the departure times, routes, and loading strategies for swap bodies, while adhering to constraints such as sorting capacities, cut-off times, and driving regulations

AUTOMATIC CONFORMANCE CHECKING INSIGHTS

Process Mining | Repository

- **Technologies:** pm4py, pycelonis, pytest, Celonis
- This project is a **"Command-line based software"** that provides automatic conformance checking insights for the Celonis data models, that allows users to choose which insights should be computed for the event logs.
- The output of the insights can be used to optimize existing process models, make conformance checking decisions and gain a better understanding of internal processes captured by the analyzed event log.

BANK MARKETING CAMPAIGN

Machine Learning | Repository

- **Technologies:** numpy, pandas, matplotlib, seaborn
- The project focused on predicting customer subscription to **"Term Deposit"** offered by a Portuguese banking institution's marketing campaigns.
- Utilizing a dataset comprising 45,211 phone calls and 20 features, the problem was approached as a **"Binary Classification"** task, aiming to determine whether customers would opt for the offering, resulting in a Yes or No decision.

ECOFORECAST: REVOLUTIONIZING GREEN ENERGY SURPLUS PREDICTION IN EUROPE

Hackathon

- We were asked to develop a model to predict the country with the highest surplus of green energy in the upcoming hour among nine specified countries. Thus, considering renewable energy generation (e.g., wind, solar, geothermal)and energy consumption to calculate the surplus.
- Ranked 13th out of 237 teams in the hackathon challenge.

B.Tech. in Computer Science and Engineering

SOA University, India CGPA: 9.83/10 (GPA: 1,1)

07/2015 – 06/2019

VEHICLE SPEED DETECTION FROM THE CAPTURED VIDEO

Computer Vision

- Developed computer-vision software to identify speeding vehicles from highway camera footage, providing automated violation detection and improving highway safety monitoring.